

RARIEDA CLUSTER EXAMINATION

Kenya Certificate of Secondary Education (K.C.S.E.)

CHEMISTRY

Paper 1

JULY 2025

MARKING SCHEME

(Confidential)

PAPER 1

233/1

1. Differentiate between fractional crystallization and fractional distillation. (2 Mks)

✓ **Fractional crystallization is the separation of two soluble salts with different solubilities in same solvent. ¹**

✓ **Fractional distillation is the separation of two miscible liquids with close boiling point. ¹**

2. Diamond and graphite are allotropes of carbon.

(i) What are allotropes. (1 Mk)

✓ **Allotropes are different forms of an element in the same physical state.**

(ii) In terms of structure and bonding explain why diamond is used in drilling through hard rocks while graphite is a lubricant. (2 Mks)

✓ **In diamond, all the four valence electrons are bounded through strong covalent bonds making it very hard. In graphite, the hexagonal layers are held together with weak van-der-walls forces which make them to slide over.**

3. Complete the table below. (3 Mks)

Metal	Aluminium	Lead	
Chief ore		Galena	
Chemical formulae	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$		NaCl
Method of extraction	Electrolysis	Reduction	Electrolysis

4. a) State Graham's law. (1 Mk)

✓ **Graham's Law states that the rate of diffusion of a gas is inversely proportional to the square root of its density at constant temperature and pressure.**

b) 60cm^3 of oxygen diffused through a porous plate in 20 seconds. How long will it take 120cm^3 of carbon (IV)oxide gas to diffuse through the same plate under the same conditions. (C=12, O=16). (2 Mks)

$$R_{\text{O}_2} = \text{cm}^3$$

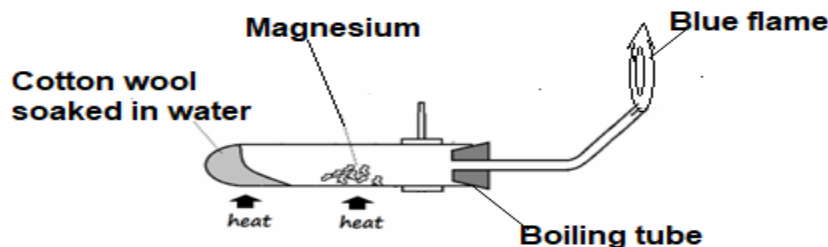
$$R_{\text{CO}_2} = 2.553$$

$$\begin{aligned} T_{\text{CO}_2} &= \\ &= 46.91\text{sec} \end{aligned}$$

5. Describe how a dry sample of barium sulphate can be prepared starting with barium carbonate. (3 Mks)

Add excess $\text{NaCO}_3(\text{s})$ to 50cm^3 of $2\text{M HNO}_3(\text{aq})$. Filter off BaCO_3 to obtain a solution of $\text{Ba}(\text{NO}_3)_2(\text{aq})$. Add Na_2SO_4 / K_2SO_4 / $(\text{nh}_4)_2\text{SO}_4$ of $\text{H}_2\text{SO}_4(\text{aq})$ to the solution add filter to obtain the residue of BaSO_4 . Rinse the residue with distilled water and dry between two filter papers.

6. Study the diagram below and answer the questions that follow.



- a) State the observations that would be made when heat is applied? (1 Mk)

White solid formed in the tube

Mg burns with a bright white flame.

- b) Write chemical equations for the reactions taking place in;

- (i) The boiling tube (1 Mk)



- (ii) The blue flame (1 Mk)



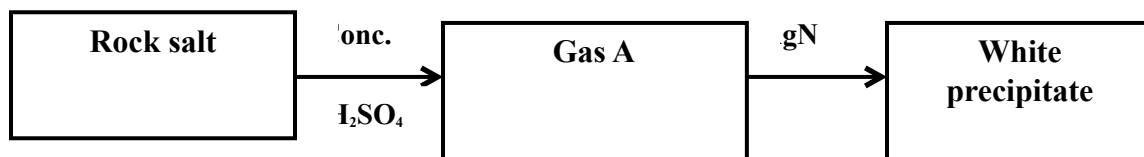
7. a) Give Browsted and Lowry definition of an acid. (1 Mk)

An acid is a proton donor.

- b) Differentiate between a strong acid and a concentrated acid. (2 Mks)

A strong acid is fully ionized in water to produce more H^+ . A concentrated acid is one which contains a higher concentration of acid

8. Study the flow chart below and use it to answer the following questions.



- a) Identify

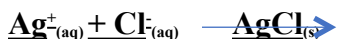
- (i) Gas A (1 Mk)

AgCl // Hydrogen chloride

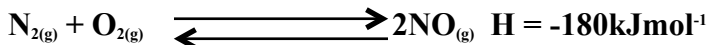
- (ii) The white precipitate (1 Mk)

$\text{Ag}_{(aq)}$ // Silver chloride

- b) Write an ionic equation for the formation of the white precipitate. (1 Mk)



9. In the production of nitrogen (II) oxide, Nitrogen and Oxygen react according to the equation.



Explain how the following would affect the yield of nitrogen (II)oxide.

- (i) Increase in pressure (1½ Mks)

Yield remains the same. Pressure has got no effect in equilibrium since number of molecules are the same.

- (ii) Increase in temperature (1½ Mks)

Yield decrease. Since the forward reaction is exothermic, increase in temperature shifts equilibrium to the left.

10. The following table shows pH values of a solution A, B and C.

Solution	A	B	C
pH	2	7	11

a) Which solution is likely to be magnesium chloride. Give a reason. (1 Mk)

B Magnesium chloride is neutral.

b) Identify the solution in which a sample of aluminium chloride is likely to dissolve. Explain (2 Mks)

C AlCl_3 hydrolyses in water to produce H^+ making it acidic.

11. The electronic configuration of ions X^{3+} and Y^{2+} are 2.8 and 2.8 respectively.

a) State the atomic numbers of each of the elements X and Y. (1 Mk)

X – 7

Y – 12

b) Write the formula of the compound formed when X and Y react. (1 Mk)

Y_3X_2 // Mg_3N_2

c) What type of bond would be formed in the compound named in (b) above. (1 Mk)

Ionic // Electrovalent

12. In an experiment a stream of dry hydrogen gas was passed over heated oxide of lead and the following results were obtained.

Mass of porcelain boat = 10.2g

Mass of porcelain boat + lead oxide = 17.37g

Final mass of porcelain boat + lead = 16.41g

Determine the formula of the oxide of lead. (Pb=207, O=16). (3 Mks)

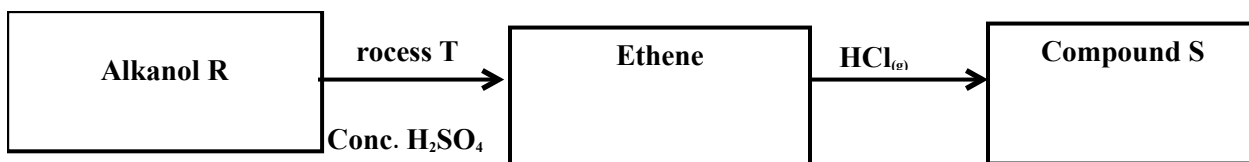
Mass of Pb = 16.41 – 10.2 = 6.21

Mass of O = 17.37 – 16.41 = 0.96

Element	Mass (g)	Mole	Ratio	Atoms
Pb	6.21			1
O	0.96			2

E.F : PbO_2

13. Study the reaction scheme below and answer the questions that follow.



a) Write the formula of

(i) Alkanol R

(1 Mk)

$\text{C}_2\text{H}_5\text{OH}$ // $\text{CH}_3\text{CH}_2\text{OH}$

(ii) Compound S (1 Mk)



b) Name process T. (1 Mk)

Dehydration

14. The following tests were carried out on chlorine water contained in two test tubes.

- a) A piece of flower petal was dropped into the first test tube. Explain why the flower petal was bleached. (2 Mks)

HOCl decomposed to produce the nascent oxygen which combined with the dye in the petal to bleach by oxidation.

- b) The second test tube was corked and it was found to contain a gas that rekindles a glowing splint. Write an equation for the reaction which produced the gas. (1 Mk)



15. State and explain how the rate of reaction between zinc granules and steam can be increased. (2 Mks)

By adding $\text{CuSO}_{4(\text{s})}$ which acts as a catalyst and hence provides a shorter path for the reaction.

16. An element Q with atomic number 31 has two isotopes as shown below.

Mass	% Abundance
69	60.4
71	39.6

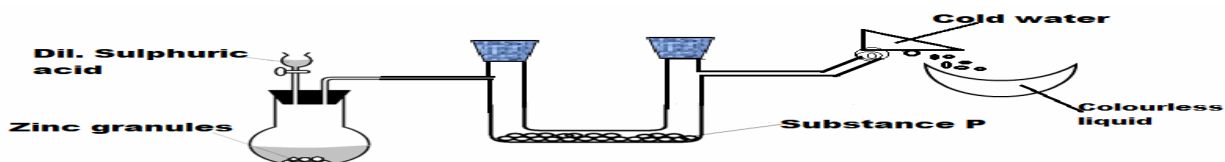
- a) Determine the number of neutrons on the isotope with mass 71. (1 Mk)

$71 - 31$
 $= 40$

- b) Calculate the relative atomic mass of element Q. (2 Mks)

+

17. Study the set up below.



- a) Name substance P. (1 Mk)

Anhydrous CaCl_2 // CaO

- b) Describe a chemical test that would be used to identify the colourless liquid. (2 Mks)

Add drops of the liquid into a test tube with anhydrous CuSO_4 . Solid changes from white to blue to confirm the liquid is water. OR Add drops of the liquid into a test tube containing anhydrous cobalt (III)chloride. Solid changes from blue to pink.

18. The table below gives some information about the physical properties of four substances L, M, N and K.

Substance	Melting point	Heat of vaporization	Electrical conductivity	
			Solid	Molten
L	High	High	Poor	Poor
M	High	High	Good	Good
N	High	High	Poor	Good
K	Low	Low	Poor	Poor

Select which elements is likely to be;

(i) Copper metal (1 Mk)

M

(ii) Silicon (IV) oxide (1 Mk)

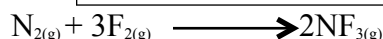
L

(iii) Potassium Iodide (1 Mk)

N

19. Use the bond energies given below to calculate the heat of formation of nitrogen tri fluoride.
(3 Mks)

Bond	N - F	N N	F - F
Bond enthalpy (kJmol ⁻¹)	278	945	159



$$\begin{aligned} \text{N} & \quad 2 \\ \text{H}_1 &= + (945 + 3(159)) \\ &= +1422 \\ \text{H}_2 &= -(6) \\ &= -1668 \end{aligned}$$

20. Dry ammonia gas was passed through a combustion tube containing heated copper (II) oxide for some time.

a) State and explain the two observations made in the combustion tube. (2 Mks)

Solid turns from black to brown, a colourless liquid settles on cooler parts. Ammonia reduces CuO to Cu but is itself oxidized to water and N₂g.

b) Name the property of ammonia gas applied in the above reaction. (1 Mk)

Reducing agent / Reducing

21. 0.24g of a divalent metal M dissolve in 50cm³ of 0.25M Sulphuric (VI) acid. The resulting solution required 5cm³ of 1M sodium hydroxide for complete neutralization. Determine the relative atomic mass of M.
(3 Mks)

$$\text{Mole of NaOH} = 0.005$$

$$\text{Mole of excess H}_2\text{SO}_{4(aq)} = \frac{1}{2}$$

$$\begin{aligned} \text{Mole of H}_2\text{SO}_4 \text{ in } 50\text{cm}^3 \text{ of } 0.25\text{M} \\ &= 0.0125 \end{aligned}$$

$$\text{Mole of H}_2\text{SO}_4 \text{ which reacted with M}$$

$$= (0.0125 - 0.0025)$$

$$= 0.01$$

$$M : H_2SO_4 = 1:1$$

$$\text{Mole of M} = 0.01$$

$$\text{R.A.M of m} =$$

$$= 24$$

22. Highlight the three stages in gas preparation. (3 Mks)

- Gas generation – main reagents react to produce the gas.
- Drying stage – suitable drying agent used to dry the gas.
- Collection stage – density of the gas used to determine the method of collection.

23. Below are properties of some elements in period 3 of the periodic table.

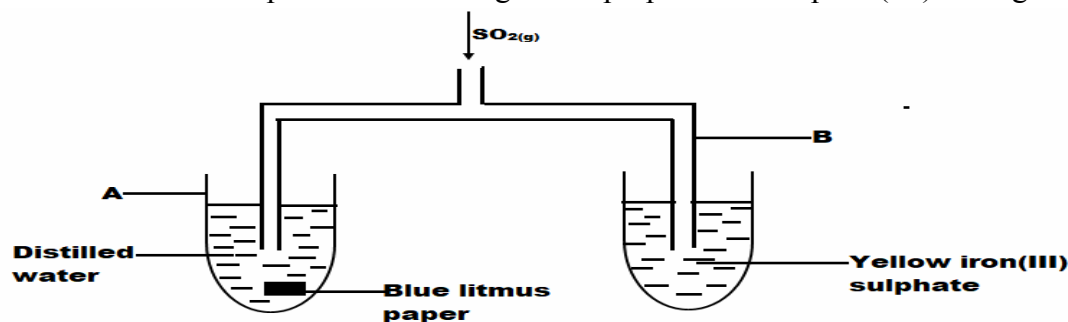
Element	Na	Mg	Al
Atomic radius	0.152	0.136	0.125
Melting points (°C)	98	650	660

Explain the trend in the melting points.

(3 Mks)

Melting points increase across the period. Increase in the number of protons leads to a decrease in atomic size which increases the strength of metallic bond.

24. Students used the set up below to investigate the properties of sulphur (IV)oxide gas.



a) State and explain the observation made in test tube A. (2 Mks)

Blue litmus paper turns red then bleached to white. $SO_{2(g)}$ dissolves in water to produce H_2SO_4 and H_2SO_3 . H_2SO_4 makes the solution acidic, while H_2SO_3 is responsible for the bleaching action.

b) What property of sulphur (IV)oxide is demonstrated by observation in test tube B.(1 Mk)

Reducing

25. a) Complete the following nuclear equations; (1 Mk)



b) A certain nuclide has a half-life of 2.5hours. What percentage of the given mass of nuclide would be left after 7.5hours? (2 Mks)

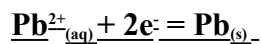
$$n =$$

$$100\% = 50\% = 25\% = 12.5\%$$

$$= 12.5\%$$

26. In the electrolysis of molten lead (II)chloride, a current of 5A was passed through the electrolyte for 386 seconds.

a) Write the ionic equation at the cathode. (1 Mk)



b) What mass of lead in grams was deposited? (Pb=207, IF=96500C). (2 Mks)

$$\underline{Q = IT = ((5 \times 386)\text{C})}$$

$$\underline{= 1930\text{C}}$$

$$\underline{2F = 207\text{g}}$$

$$\underline{2 \times 96500\text{c} = 207\text{g}}$$

=

27. Hydrogen chloride gas was passed through pure water and through methyl benzene in separate test tubes. Blue litmus paper was then dipped in each of the solution formed. State and explain the observation made in each case.

a) In pure water (1½ Mks)

Blue litmus paper turns to red $\text{HCl}_{(\text{g})}$ hydrolyses in water to produce H^+ making solution acidic.

b) In methylbenzene (1½ Mks)

Blue litmus paper remains blue. $\text{HCl}_{(\text{g})}$ dissolves in methylbenzene but remains as a molecule. // does not produce H^+ .